PROJECT REPORT

US Census Income Prediction

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Introduction

The objective of this Project is to apply exploratory data analysis and predictive modeling techniques to a real-world dataset in order to examine determinants of individual income. Using the American Community Survey (ACS) Public Use Microdata Sample (PUMS), this project investigates whether demographic and employment-related characteristics can be used to predict total personal income (PINCP).

The primary research question guiding this analysis is:

Can an individual's total personal income be predicted using demographic, educational, and employment-related characteristics?

To answer this question, exploratory data analysis (EDA) was performed to understand distributions and relationships among variables. Subsequently, three predictive models - Linear Regression, Random Forest Regression, and Gradient Boosting Regression were developed and compared to evaluate predictive performance.

Dataset Description & Data Extraction

The dataset used in this study is derived from the American Community Survey (ACS) Public Use Microdata Sample (PUMS), publicly available from the U.S. Census Bureau (https://www.census.gov/programs-surveys/acs/microdata.html).

The dataset contains approximately 33,000 individual-level observations and 290 variables. Each record represents a person and includes demographic attributes, employment status, income measures, and related socioeconomic indicators.

```
data.head()
```

[33]:

	PINCP	AGEP	SEX	MAR	ENG	LANX	COW	ESR	WKHP	JWMNP
0	0.0	36	1	5	NaN	2.0	1.0	3.0	NaN	NaN
1	30000.0	19	2	5	1.0	1.0	5.0	4.0	40.0	5.0
2	21600.0	23	1	5	NaN	2.0	1.0	1.0	77.0	NaN
3	6300.0	19	1	5	1.0	1.0	1.0	6.0	70.0	NaN
4	3500.0	33	1	3	NaN	2.0	3.0	6.0	40.0	NaN

For this analysis, a subset of relevant variables was selected:

- PINCP (Total Personal Income) – Target variable
- AGEP (Age)
- SEX (Gender)

- MAR (Marital Status)
- ENG (English proficiency)
- LANX (Language spoken at home)
- COW (Class of worker)
- ESR (Employment status)
- WKHP (Hours worked per week)
- JWMNP (Commute time)

This subset supports investigation of income determinants while maintaining interpretability.

Tools and Packages Used

The analysis was conducted using Python. The following libraries were used:

- Pandas – Data manipulation
- NumPy – Numerical operations
- Matplotlib & Seaborn – Data visualization
- Scikit-learn – Predictive modeling
- Jupyter Notebook – Analysis workflow

A separate Python script file (.py) containing three user-defined functions was created for modularization:

1. Dataset loading
2. Data cleaning
3. Feature-target splitting

The Jupyter Notebook imported these functions and executed the main analysis logic.

Data Cleanup

```
data.isnull().sum()
```

```
PINCP      6755
AGEP          0
SEX           0
MAR           0
ENG       27688
LANX       2090
COW       13191
ESR        7219
WKHP      15723
JWMNP     19916
dtype: int64
```

Data preparation involved several steps:

1. Selection of relevant columns for modeling
2. Removal of missing values in the target variable (PINCP)
3. Replacement of missing predictor values with zero
4. Verification that no null values remained
5. Separation of predictors (X) and target (y)

```
data = data.dropna(subset=["PINCP"])
```

```
data.isnull().sum()
```

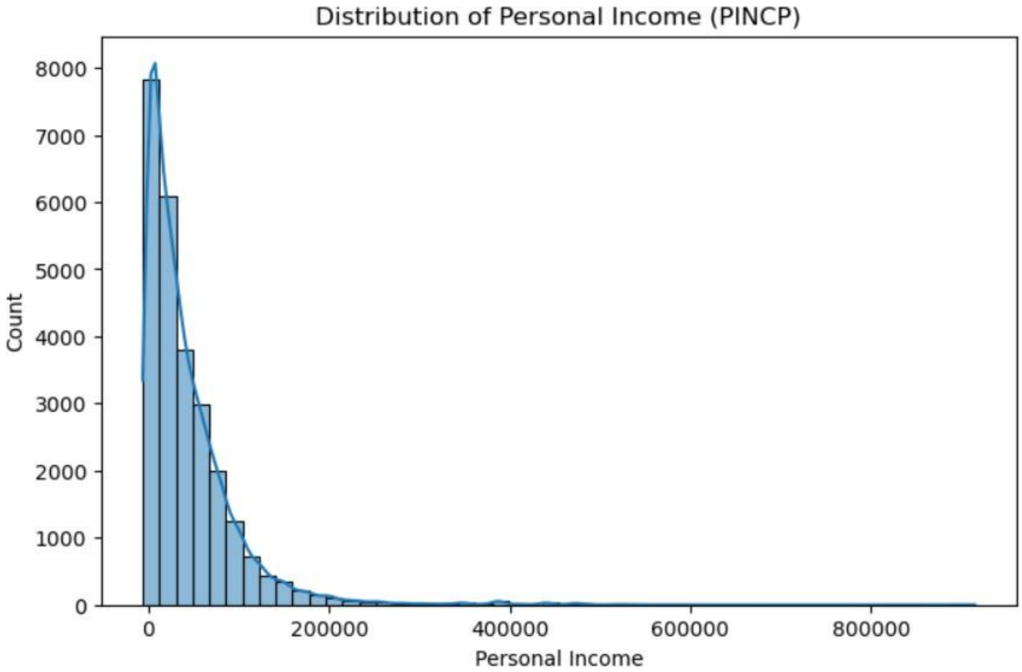
```
PINCP        0
AGEP         0
SEX          0
MAR          0
ENG      21576
LANX         0
COW       6436
ESR        464
WKHP      8968
JWMNP    13161
dtype: int64
```

Initial inspection revealed missing values in ENG, COW, WKHP, and JWMNP. After applying cleaning procedures, all missing values were resolved, ensuring suitability for modeling.

Exploratory Data Analysis

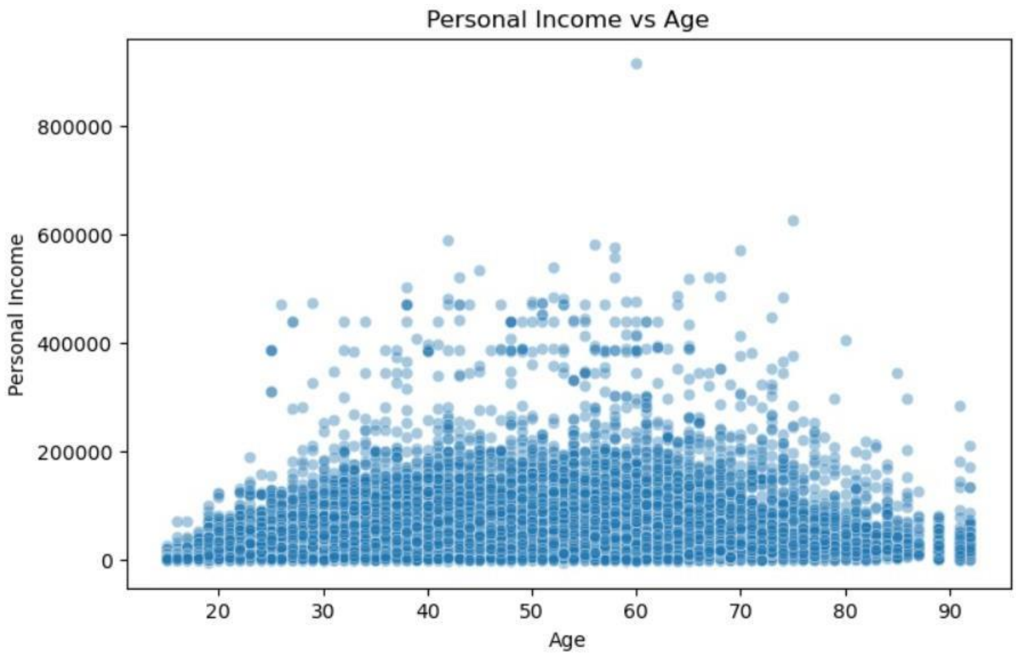
Six visualizations were created to explore key relationships.

1. Distribution of Personal Income



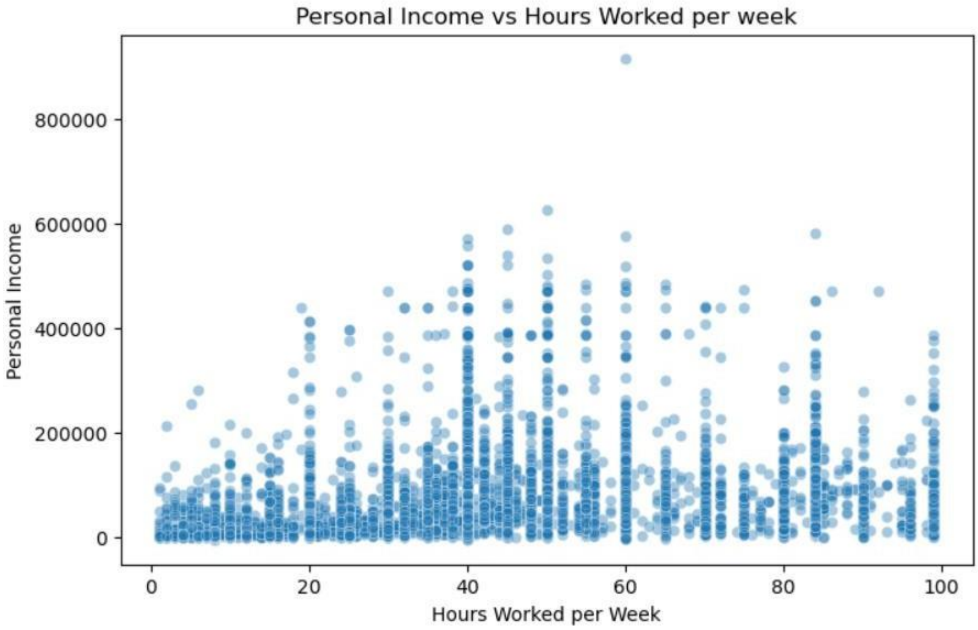
The histogram reveals a strongly right-skewed distribution. Most individuals earn lower to moderate incomes, while a small proportion earn substantially higher incomes. This skewness supports the need for regression-based modeling techniques capable of handling variance.

2. Personal Income vs Age



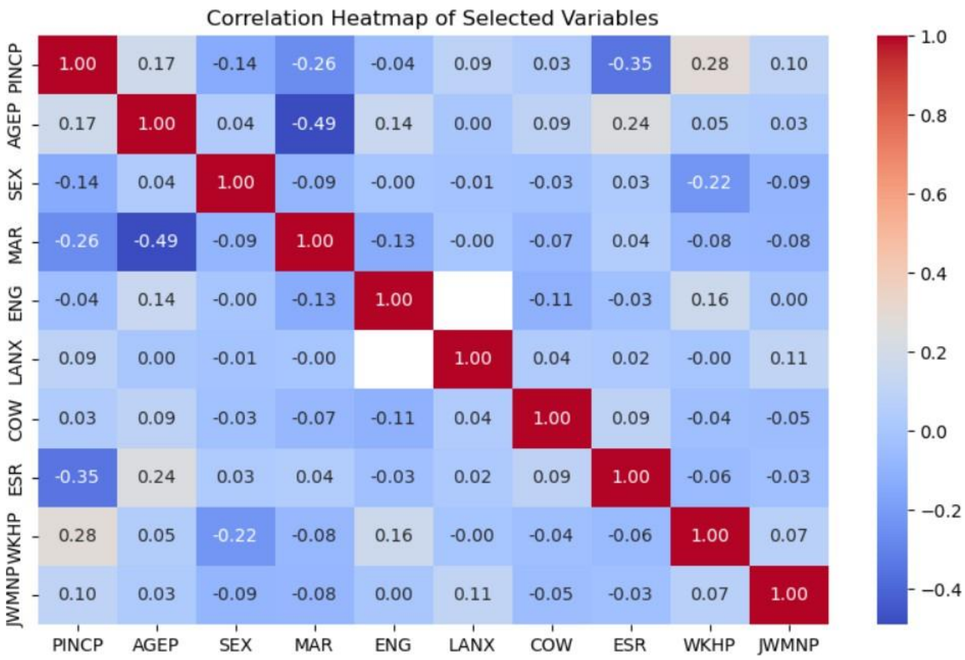
A positive relationship between age and income is observed. Income generally increases with age, likely reflecting accumulated work experience. However, wide dispersion indicates age alone does not fully explain income variability.

3. Personal Income vs Weekly Hours Worked



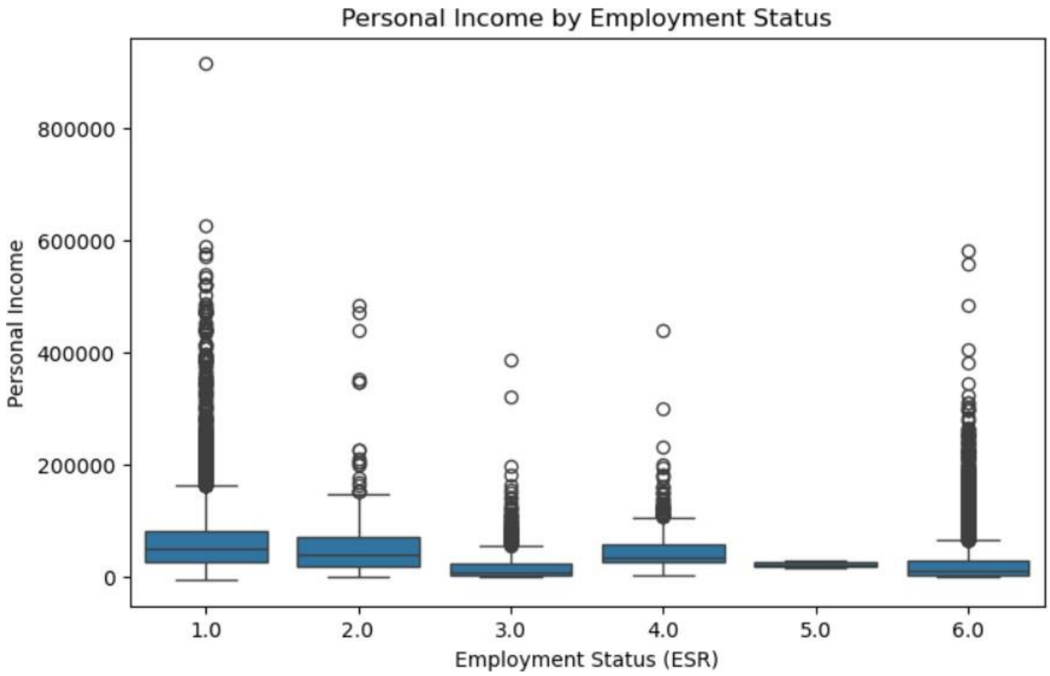
The scatter plot shows that individuals working more hours per week tend to earn higher incomes. Nevertheless, variability suggests additional demographic and employment characteristics influence income levels.

4. Correlation Matrix



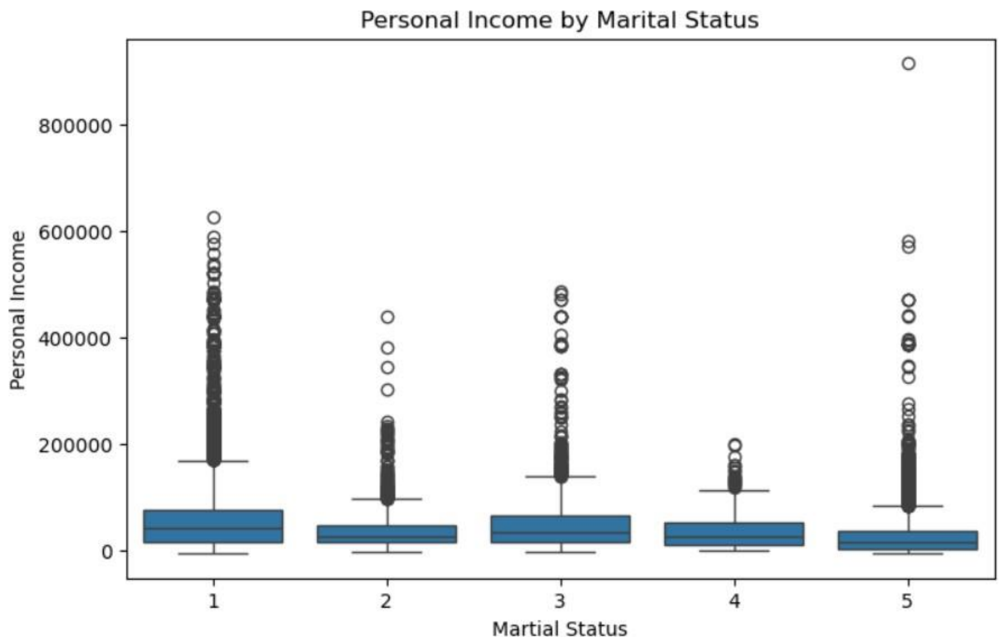
The heatmap shows moderate positive correlations between PINCP and AGEP (0.17) and WKHP (0.28). No severe multicollinearity is observed, supporting inclusion of all selected predictors in regression models.

5. Income by Employment Status



Boxplots indicate that employed individuals earn substantially more than those unemployed or not in the labor force. Employment status appears to be a strong predictor.

6. Income by Marital Status



Income distributions vary across marital categories, suggesting household structure may influence economic outcomes.

Predictive Modeling

To answer the research question whether personal income can be predicted using demographic and employment-related features, three regression models were developed and evaluated: Linear Regression, Random Forest Regression, and Gradient Boosting Regression.

The dataset was split into training (80%) and testing (20%) subsets using train_test_split with a fixed random state (42) to ensure reproducibility. The models were trained on the training data and evaluated on the test data using three performance metrics:

- Mean Absolute Error (MAE) – Average absolute prediction error
- Root Mean Squared Error (RMSE) – Penalizes larger errors more heavily
- R^2 (Coefficient of Determination) – Proportion of variance explained

Using multiple evaluation metrics provides a comprehensive understanding of predictive performance.

Model 1: Linear Regression

Linear Regression was selected as a baseline model due to its interpretability and its widespread use in economic research. This model assumes a linear relationship between predictors and income, meaning that changes in variables such as age, hours worked, or employment status have additive and proportional effects on income. The model was fitted using the selected demographic and employment variables.

```
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

# Predictions
y_pred_lr = model_lr.predict(X_test)

# Evaluation metrics
mae = mean_absolute_error(y_test, y_pred_lr)
rmse = np.sqrt(mean_squared_error(y_test, y_pred_lr))
r2 = r2_score(y_test, y_pred_lr)

mae, rmse, r2
```

```
(27077.496733209035, np.float64(47346.38199017367), 0.27899162061438887)
```

Results:

- MAE: 27,077

- RMSE: 47,346
- R^2: 0.279

An R^2 value of 0.279 indicates that approximately 28% of the variation in personal income can be explained by the selected predictors under a linear assumption. While the model captures some income variability, the relatively high RMSE suggests substantial prediction error. This outcome reflects the complexity of income determination, which may involve non-linear interactions and structural effects not captured by a purely linear framework.

Thus, Linear Regression provides a useful benchmark but demonstrates limitations in modeling income dynamics.

Model 2: Random Forest Regression

Random Forest Regression was implemented to address limitations of linear modeling. Unlike Linear Regression, Random Forest does not assume linear relationships. Instead, it constructs multiple decision trees using bootstrapped samples of the data and aggregates their predictions.

This ensemble approach:

- Captures non-linear relationships
- Handles interactions between variables
- Reduces overfitting through averaging
- Works effectively with mixed variable types

```
y_pred_rf = model_rf.predict(X_test)

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

mae_rf = mean_absolute_error(y_test, y_pred_rf)
rmse_rf = np.sqrt(mean_squared_error(y_test, y_pred_rf))
r2_rf = r2_score(y_test, y_pred_rf)

mae_rf, rmse_rf, r2_rf
```

```
(25314.330776491617, np.float64(47054.86779616975), 0.2878428616257438)
```

Results:

- MAE: 25,314
- RMSE: 47,054
- R^2: 0.288

Compared to Linear Regression, Random Forest reduced MAE by approximately 1,800 units and slightly improved R^2 to 0.291. This improvement suggests that income relationships are not purely linear and that tree-based models better capture threshold effects (e.g., employment category shifts, age brackets). However, the performance gain is modest, indicating that while non-linearity matters, there may still be structural limitations in the available predictors.

Model 3: Gradient Boosting Regression

Gradient Boosting Regression was implemented as the most advanced modeling technique in this study. Unlike Random Forest, which builds trees independently, Gradient Boosting constructs trees sequentially. Each subsequent tree focuses on correcting errors made by the previous trees.

```
y_pred_gb = model_gb.predict(X_test)

from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_score
import numpy as np

mae_gb = mean_absolute_error(y_test, y_pred_gb)
rmse_gb = np.sqrt(mean_squared_error(y_test, y_pred_gb))
r2_gb = r2_score(y_test, y_pred_gb)

mae_gb, rmse_gb, r2_gb
```

```
(24116.144278218475, np.float64(45172.67064852808), 0.34367605205471663)
```

Results:

- MAE: 24,116
- RMSE: 45,172
- R^2: 0.344

Gradient Boosting achieved the highest predictive performance, explaining approximately 34.4% of income variation.

The reduction in MAE (nearly 3,000 units lower than Linear Regression) and the higher R^2 indicate that boosting effectively captures more complex relationships between demographic and employment characteristics and income levels.

This performance confirms that income determination involves layered and interacting effects that sequential error correction models are better suited to capture.

Model Comparison

Model	MAE	RMSE	R^2
Linear Regression	27,077	47,346	0.279
Random Forest	25,314	47,054	0.288
Gradient Boosting	24,116	45,172	0.344

The progression of results demonstrates a clear pattern:

- Linear model provides baseline explanatory power.
- Random Forest improves performance by capturing non-linearities.
- Gradient Boosting further improves predictive accuracy through iterative error correction.

Despite improvements, even the best model explains approximately 34% of income variability. This indicates that income is influenced by additional unobserved factors such as occupation type, industry, geographic effects, education level depth, economic conditions, or structural labor market dynamics.

Thus, while demographic and employment variables provide meaningful predictive power, they do not fully determine income outcomes.

Interpretation & Conclusion

The analysis demonstrates that demographic and employment-related characteristics can moderately predict individual income. The best model (Gradient Boosting) explains approximately 34% of income variation.

Key findings include:

- Age and hours worked are positively associated with income.
- Employment status significantly differentiates income levels.
- Non-linear models outperform linear regression.

Although prediction accuracy is moderate, results confirm that income is influenced by multiple interacting factors. Advanced ensemble methods better capture these relationships.

In conclusion, demographic and employment characteristics provide meaningful, though incomplete, predictive power for estimating personal income. These findings can inform workforce planning, economic policy analysis, and labor market research.

References

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